

Homework set 4

1) Poincaré algebra:

Consider the Poincaré group (Λ^μ_ν, a^μ)

in some representation $S(\Lambda, a)$.

Expand the represent. matrices around

δ^μ_ν , and derive the Poincaré algebra

by imposing multiplication rule for the Poincaré group.

2) Symmetries of Euclidean field theory

Consider Euclidean flat space $\mathbb{R}^{d+1}$.

Work out the commutators of the Lie algebra of $SO(d+1)$. Identify the generators of translations & rotations.

3) proper length

In 3+1 dim a Lorentz vector is in the irrep $(\frac{1}{2}, \frac{1}{2})$. Write down its proper length in terms of the spinor d.o.f.

4) Show that $SU(2)$ is a double cover of $SO(3)$.

5) Clifford algebra:

Show that $\sigma_{\alpha i}^{\mu}$ & $\bar{\sigma}_{\alpha i}^{\mu}$ satisfy the Clifford algebra

$$(\sigma_{\alpha}^{\mu} \bar{\sigma}_{\alpha}^{\nu} + \sigma_{\alpha}^{\nu} \bar{\sigma}_{\alpha}^{\mu})^{\beta} = 2\eta^{\mu\nu} \delta_{\alpha}^{\beta}$$

$$(\bar{\sigma}^{\mu} \sigma^{\nu} + \bar{\sigma}^{\nu} \sigma^{\mu})^{\dot{\alpha}} = 2\eta^{\mu\nu} \delta^{\dot{\alpha}}$$

6) P^2 is a Lorentz-invariant quantity and commutes with all Poincaré generators (it is a quadratic Casimir). The 3+1 dim Poincaré algebra has another quadratic casimir:

$$W^2 = W^\mu W_\mu \quad \text{where}$$

$$W^\mu = \frac{1}{2} \epsilon^{\mu\nu\alpha\beta} P_\nu J_{\alpha\beta}$$

Show that

$$W^2 = \frac{1}{2} P^2 J^{\mu\nu} J_{\mu\nu} + P_\mu J^{\mu\nu} P_\nu J^\mu_\nu$$

Apply this to massless particle states & write it in terms of A , B and J . (the generators we encountered in the little group of massless particles)